

Eric Waters

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Software Engineer with an M.S. in Big Data Systems and 3+ years of experience at Intel and Dell. Skilled in both backend and frontend, with extensive experience in building APIs, responsive UIs, data pipelines, AI/ML integrations, and cloud/DevOps solutions.

Experience

Intel Corporation

Chandler, Arizona

Software Application Development Engineer

August 2023 – Present

Offline Data Analytics Platform

- Architected and launched a centralized Offline Analytics Platform, integrating Airflow, Kubernetes, CI/CD, and ELK to standardize ETL development and eliminate duplicate tooling across teams.
- Reduced ETL onboarding time from 2 weeks to 2 days by delivering reusable pipelines, preconfigured infrastructure, and Python libraries for rapid DAG development and logging.

Generative AI Web Platform

- Developed a suite of 5 GenAI applications using Angular and Python (Sanic), including a legal contract audit tool, estimated to save \$30M annually (recipient of a Divisional Recognition Award).
- Architected and designed a no-code drag-and-drop AI app studio using React, Next.js, Tailwind CSS, enabling non-technical teams to build custom AI workflows efficiently.
- Redesigned the platform's UI/UX, implementing reusable frontend components to standardize design language and accelerate development cycles across 6+ developers.

Factory Excursion Management API

- Spearheaded the architectural merger of two massive FastAPI instances, unifying authentication (OAuth 2.0/OpenID), logging, and CI/CD pipelines to reduce long-term maintenance overhead.

Factory ML Analysis API

- Engineered a high-availability FastAPI service to expose legacy .NET ML algorithms to external applications, handling hundreds of thousands of requests per month.

AI Cognitive Search Engine

- Sole Administrator for a cross-company cognitive search engine; developed ETL pipelines using Python, Pandas, and SQL Server to index millions of documents from SharePoint, Wikis, and other internal manufacturing document sources.

Dell Technologies

Round Rock, Texas

Software Engineering Intern (Undergraduate & Graduate)

May – August 2021 & June – August 2022

- Implemented OAuth 2.0 SSO and fine-grained authorization across enterprise apps and APIs using Spring Boot and Angular, enhancing security, compliance, and user experience.
- Built a custom Kanban board app with Spring and MySQL, improving agile collaboration for engineering teams.

Education

Arizona State University

Tempe, Arizona

Computer Science (Big Data Systems), M.S. with Distinction

4.0 GPA | May 2023

Computer Science, B.S. Summa Cum Laude & Barrett, the Honors College

4.0 GPA | May 2022

Skills

- **Programming Languages:** Python, Java, TypeScript, JavaScript, C/C++, SQL
- **Frontend:** React, Next.js, Angular, Tailwind CSS
- **Backend:** FastAPI, Sanic, Spring Boot, REST, OAuth 2.0
- **Data & ETL:** Pandas, Airflow, Kafka, Spark, PostgreSQL, SQL Server, MySQL
- **DevOps & Cloud:** Docker, Kubernetes, Rancher, GitHub Actions, ELK Stack
- **AI/ML:** TensorFlow, Scikit-learn, Keras

Projects

Automated Detection of Phishing Attacks using Machine Learning

March 2023 – May 2023

- Evaluated 9 machine learning techniques against a dataset of legitimate and phishing URLs using Scikit-learn and Pandas
- Assessed the performances of PCA, Kernel PCA, and autoencoder dimensionality reduction techniques
- Achieved 98% accuracy by training an ensemble classifier of optimized ML techniques and deep neural networks

Convolutional Neural Network (Image Classifier)

May 2022

- Designed neural networks using Python and TensorFlow to classify handwritten digits (trained with MNIST dataset) with 99% accuracy and images from 10 categories (trained with CIFAR-10 dataset) with 81% accuracy